

Phase-Based Gravitational Dynamics and Synthetic Metrics: An Epistemic Analysis of Quantum Interferometry and Spatio-Temporal Modulation in the Context of UAT/UPC

Miguel A. Percudani

Independent Researcher, Argentina

ORCID: 0009-0007-1748-3212

September 24, 2026

Abstract

Recent experimental developments in quantum interferometry and spatio-temporally modulated resonant systems provide important empirical context for phase-based approaches to physical dynamics. In particular, the direct measurement of quantum phase accumulation during free fall reported by Dobkowski et al. [6], and the experimental realization of magnetic-free non-reciprocity through parametric phase modulation reported by Estep, Souнас, Soric, and Alù [5], demonstrate that phase evolution and its controlled spatio-temporal modulation can constitute experimentally observable physical degrees of freedom.

These results do *not* validate the Universal Applied Time (UAT) [1] or Unified Principle of Causality (UPC) [2] frameworks. Both experiments were developed and interpreted within established physical frameworks and address specific experimental questions unrelated to the validation of UAT/UPC. Nevertheless, they provide relevant experimental precedents for a research strategy in which discrete phase geometry, temporal modulation, and coherent superposition are treated as primary experimental variables.

Within this context, the present work identifies the eight-coil rotational matrix developed under the UAT/UPC program [4] as an exploratory macroscopic platform for testing whether controlled phase geometry can produce reproducible collective observables. The objective is not to assume the existence of a new metric or causal field, but to determine experimentally whether a reproducible residual or convergence parameter emerges from an independently calibrated phase-rotational system.

Keywords: quantum phase, free-fall interferometry, spatio-temporal modulation, synthetic gauge field, eight-coil detector

1 Introduction

The experimental investigation of gravitational phenomena at the quantum level increasingly emphasizes the role of phase as an observable quantity. A particularly relevant recent result is the direct measurement of the quantum phase associated with free fall reported by Dobkowski et al. [6]. Their experiment employs a Quantum Galileo Interferometer (QGI) to measure the differential phase accumulated between a stationary and a freely falling component of an atomic wave packet.

The observed time dependence contains a cubic contribution $\phi_{\text{ff}} \propto T^3$, consistent with theoretical expectations.

This result should not be interpreted as evidence for UAT or UPC. Rather, it establishes an experimentally demonstrated example in which gravitational dynamics are directly encoded in a measurable quantum phase.

$$\boxed{\text{experimental evidence for gravitational phase evolution} \neq \text{experimental validation of UAT/UPC}} \quad (1)$$

2 The Alù Precedent: Spatio-Temporal Phase Modulation

A second relevant precedent is provided by Estep, Sounas, Soric, and Alù [5], who demonstrated magnetic-free non-reciprocity in a system consisting of three strongly coupled resonators whose resonance frequencies were parametrically modulated with phase shifts of $\Delta\phi = 2\pi/3 = 120^\circ$.

The result establishes a general principle: the relative temporal phase of spatially distributed resonant elements can modify the collective dynamical response.

$$\text{distributed resonators} + \text{temporal phase modulation} \longrightarrow \text{modified collective dynamics.} \quad (2)$$

The Alù experiment does not demonstrate gravitational dynamics, synthetic space-time curvature, UAT, or UPC. Its relevance is that it independently demonstrates that phase geometry can be treated as an active experimental variable.

Crucially, as noted in prior analysis, the coils do not need to be physically opposed to create a rotational geometry. The geometry can be encoded in phase/frequency relationships:

$$\phi_k(t) = \phi_0 + k\Delta\phi(t) \quad (3)$$

3 From Three-Element Phase Modulation to an Eight-Coil Matrix

For an ideal eight-element rotational phase structure, the discrete phase increment is

$$\Delta\phi = \frac{2\pi}{8} = \frac{\pi}{4} = 45^\circ, \quad \phi_k = \phi_0 + k\frac{\pi}{4}, \quad k = 0, \dots, 7. \quad (4)$$

Thus,

$$\boxed{\text{physical coil array} + \text{controlled phase relationships} = \text{synthetic rotational configuration}}. \quad (5)$$

The eight-coil detector, calibration code, and methodology are documented in Percudani [4].

4 The Eight-Coil Rotational Matrix

The system uses eight phase fronts $\{0^\circ, 45^\circ, 90^\circ, 135^\circ, 180^\circ, 225^\circ, 270^\circ, 315^\circ\}$. The number of unordered pairs is $\binom{8}{2} = 28$, noted here only as a structural property.

5 Rotational Superposition and Residual Signal

For an idealized set of eight equal-amplitude phase components:

$$S_8 = \sum_{k=0}^7 A e^{i(\phi_0 + k\pi/4)} = A e^{i\phi_0} \sum_{k=0}^7 e^{ik\pi/4}. \quad (6)$$

Since $e^{i8\pi/4} = e^{i2\pi} = 1$, for a perfectly balanced set:

$$\boxed{S_8 = 0}. \quad (7)$$

Therefore any experimentally measured residual $S_{\text{res}} \neq 0$ can be decomposed as:

$$S_{\text{res}} = S_{\text{instrumental}} + S_{\text{phase}} + S_{\text{frequency}} + S_{\text{electronic}} + S_{\text{physical}}. \quad (8)$$

where

$$S_{\text{res}} = S_{\text{ideal}} + S_{\text{residual}}, \quad S_{\text{ideal}} = 0 \implies S = S_{\text{res}}. \quad (9)$$

6 Instrumental vs Structural Residual

Following the prior conceptual analysis, two possibilities must be distinguished:

A. Instrumental residual:

$$R = R_{\text{coils}} + R_{\text{phase}} + R_{\text{frequency}} + R_{\text{ADC}} + R_{\text{amplifier}} + R_{\text{noise}}. \quad (10)$$

In this case 0.2971 is a characteristic of the apparatus.

B. Structural residual:

$$R_{\text{observed}} = R_{\text{instrumental}} + R_{\text{structural}}, \quad (11)$$

and after characterization $R_{\text{structural}} \approx 0.2971$. Only in case B would it be meaningful to investigate 0.2971 as a candidate invariant.

7 The Observed Residual and the Candidate Invariant

During development, a residual close to

$$\boxed{R \simeq 0.2971} \quad (12)$$

was identified under phase/frequency confrontation.

This value is treated here as a *candidate invariant*, not as an established universal constant. Before assigning physical meaning, the exact mathematical operator must be defined:

$$R = \frac{A_{\text{residual}}}{A_{\text{total}}} \quad \text{vs} \quad R = \sqrt{\frac{\sum x_{\text{residual}}^2}{\sum x_{\text{total}}^2}} \quad \text{vs} \quad R = |S| \quad \text{vs} \quad R = \frac{RMS_{\text{residual}}}{RMS_{\text{reference}}}. \quad (13)$$

The scientific value depends on the operator choice.

A suitable program measures $R = R(f, A, \phi_0, \Delta\phi, T, G, \dots)$.

8 The RMS Coherence Scale

The calibration study identified $RMS_0 \simeq 0.707001 \approx 1/\sqrt{2} = 0.70710678\dots$ [4]. This quantity and $R \simeq 0.2971$ must not be assumed to represent the same observable without defining the generating operator.

9 Proposed Invariance Tests [Incorporating ChatGPT advice]

To determine whether 0.2971 is (i) instrumental noise, (ii) phase/frequency imperfection, (iii) mathematical property of the 8-channel transform, or (iv) physical residual, the following protocol is proposed:

Test 1 – Ideal signals: $x_k(t) = A \cos(\omega t + k\pi/4)$ should yield $R = 0$, verifying the algorithm.

Test 2 – Real channels: Same operator applied to measured $x_1(t), \dots, x_8(t)$ yields R_{real} .

Test 3 – Permutation: Random permutation of order 1..8. If R disappears, dependence on phase order exists.

Test 4 – Phase inversion: $\phi_k \rightarrow -\phi_k$. Instrumental vs structural residuals respond differently.

Test 5 – Frequency variation: $R(f)$ around working frequency f_0 . A peak $R(f_0) \approx 0.2971$ suggests frequency structure.

Decisive test – Parameter invariance: Varying $f_0 \rightarrow f_1$, $A \rightarrow A'$, $\Delta\phi \rightarrow \Delta\phi'$, integration time $\tau \rightarrow \tau'$, gain, bandwidth. If $R_1 = 0.2970$, $R_2 = 0.2972$, $R_3 = 0.2971$ under distinct conditions, with protocol fixed *a priori* (not tuned to obtain 0.2971), the result acquires weight as experimental attractor.

10 Conceptual Convergence

$$\begin{aligned} \text{Dobkowski et al.} &\rightarrow \text{direct measurement of gravitational quantum phase,} \\ \text{Estep et al./Alù} &\rightarrow \text{controlled spatio-temporal phase modulation,} \\ \text{8-coil UAT/UPC experiment [4]} &\rightarrow \text{test of collective rotational phase coherence.} \end{aligned} \tag{14}$$

Accordingly,

$$\boxed{\text{Dobkowski et al. and Alù et al. do not validate UAT/UPC}} \tag{15}$$

11 Epistemic Boundaries and Falsifiability

1. Quantum gravitational phase accumulation has been experimentally measured in an established framework.
2. Spatio-temporal phase modulation has experimentally produced non-reciprocal dynamics.
3. The eight-coil configuration provides a testable implementation of discrete phase relationships.
4. A residual near 0.2971 has been observed within the corresponding analysis.
5. The physical universality of 0.2971 remains an open experimental question.
6. None of the preceding results constitutes direct validation of UAT or UPC.

12 Experimental Objective

$$H_0 : R \neq 0 \text{ is entirely explained by instrumental effects,} \tag{16}$$

$$H_1 : R \approx 0.2971 \text{ persists after instrumental effects are removed.} \tag{17}$$

If the residual disappears under shielding, independent instrumentation, phase randomization, or reversal, its origin is systematic. If stable, further investigation is justified.

13 Conclusion

The direct measurement of gravitational quantum phase by Dobkowski et al. and the demonstration of collective dynamics via phase modulation by Alù et al. establish that phase geometry and temporal modulation can be treated as legitimate experimental variables. Neither validates UAT/UPC. The eight-coil rotational matrix [4] represents a distinct attempt to investigate whether a related phase-structured architecture produces a reproducible collective residual. The candidate value $R \simeq 0.2971$ remains an empirical hypothesis whose significance depends on independent reproduction, survival under controls, and invariance under parameter changes.

References

- [1] M. A. Percudani, *Universal Applied Time (UAT): A Causal Framework for Rotational Coherence*, Zenodo (2025). DOI: <https://doi.org/10.5281/zenodo.17729221>
- [2] M. A. Percudani, *Unified Principle of Causality (UPC): Multiscale Homeostasis and the Bit of Authority*, Zenodo (2025). DOI: <https://doi.org/10.5281/zenodo.18210808>
- [3] M. A. Percudani and J. I. Díaz, *Resonant Hunter v8.4 – Gravitational Lens Signal Analysis*, Zenodo (2025). DOI: <https://doi.org/10.5281/zenodo.18446712>
- [4] M. A. Percudani, *Calibration Code and Technical Report for an 8+1 Coil Rotational Detector Based on the UAT/UPC Frameworks*, Zenodo (2026). DOI: <https://doi.org/10.5281/zenodo.19646349>
- [5] N. A. Estep, D. L. Souнас, J. Soric, and A. Alù, Magnetic-free non-reciprocity and isolation based on parametrically modulated coupled-resonator loops, *Nature Physics*, vol. 10, pp. 923–927 (2014). DOI: <https://doi.org/10.1038/nphys3134>
- [6] O. Dobkowski, R. Piccon, T. P. Zibrova, A. Baranov, S. De, S. H. Cantu, E. M. Rasel, D. Rohrlisch, and C. Henkel, Direct measurement of the quantum phase of free fall, *Science Advances*, vol. 12, no. 3, eadj8028 (2026). DOI: <https://doi.org/10.1126/sciadv.adj8028>